This article was downloaded by: [University of Iowa Libraries]
On: 14 March 2015, At: 06:35
Publisher: Routledge
Informa Ltd Registered in England and Wales Registered Number: 1072954
Registered office: Mortimer House, 37-41 Mortimer Street, London W1T 3JH, UK



European Journal of Science Education

Publication details, including instructions for authors and subscription information:
http://www.tandfonline.com/loi/tsed19

Conceptual change and science teaching

Kenneth A. Strike [a] & George J. Posner [a]

[a] Department of Education, New York State College of Agriculture and Life Sciences , Ithaca, New York 14853, USA

Published online: 24 Feb 2007.

To cite this article: Kenneth A. Strike & George J. Posner (1982) Conceptual change and science teaching, European Journal of Science Education, 4:3, 231-240, DOI: 10.1080/0140528820040302

To link to this article: http://dx.doi.org/10.1080/0140528820040302

PLEASE SCROLL DOWN FOR ARTICLE

Taylor & Francis makes every effort to ensure the accuracy of all the information (the "Content") contained in the publications on our platform. However, Taylor & Francis, our agents, and our licensors make no representations or warranties whatsoever as to the accuracy, completeness, or suitability for any purpose of the Content. Any opinions and views expressed in this publication are the opinions and views of the authors, and are not the views of or endorsed by Taylor & Francis. The accuracy of the Content should not be relied upon and should be independently verified with primary sources of information. Taylor and Francis shall not be liable for any losses, actions, claims, proceedings, demands, costs, expenses, damages, and other liabilities whatsoever or howsoever caused arising directly or indirectly in connection with, in relation to or arising out of the use of the Content.

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or distribution in any form to anyone is expressly forbidden. Terms & Conditions of access and use can be found at http://www.tandfonline.com/page/terms-and-conditions

EUR. J. SCI. EDUC., 1982, VOL. 4, NO. 3, 231–240

Conceptual change and science teaching

Kenneth A. Strike and George J. Posner, Department of Education, New York State College of Agriculture and Life Sciences, Ithaca, New York 14853, USA

Downloaded by [University of Iowa Libraries] at 06:35 14 March 2015

Summaries

English
In this article, the authors address themselves to the important question of how learning in science education relates to the growth of scientific knowledge. After discussing how changes in scientific concepts come about, they offer suggestions for modifying the objectives and content of science curricula and teaching approaches in science education in the light of knowledge and understanding of the history and philosophy of science.

Deutsch
In diesem Beitrag widmen sich die Autoren der wichtigen Frage, in welcher Weise das Lernen im naturwissenschaftlichen Unterricht in Beziehung zum Anwachsen des naturwissenschaftlichen Wissens steht. Nach einer Diskussion darüber, wie Änderungen in naturwissenschaftlichen Begriffen zustandekamen, bieten sie Vorschläge dafür an, wie man die Ziele und Inhalte von naturwissenschaftlichen Curricula und Entwürfen für den naturwissenschaftlichen Unterricht ändern kann, wenn man die Kenntnis und das Verständnis der Geschichte der Naturwissenschaften mit einbezieht.

Français
Dans ce rapport, les auteurs se consacrent au problème très important du rapport entre l'étude des sciences naturelles et l'accroissement des connaissances scientifiques. A la suite d'une discussion sur la manière dont surviennent des changements de concepts en sciences naturelles, ils présentent une possibilité de changer les objectifs et le contenu des curricula scientifiques et l'approche éducative de l'enseignement des sciences naturelles, à l'aide de connaissances et de la compréhension de l'histoire des sciences.

What is learning? What is the process by which scientific knowledge grows? How does learning in science education relate to the growth of scientific knowledge? In this article we address these questions, concluding that both the learning of science and the growth of scientific knowledge must be understood as more than the accumulation of facts; that they must both be characterized as the transformation of current knowledge; that understanding both learning and the history of science requires us to understand how conceptions change. After discussing views of how changes in scientific concepts have come about, we offer suggestions for modifying curricular

0140 5284/82/0403 0231 $02·00 © 1982 Taylor & Francis Ltd

objectives, content, and teaching strategies in science from what we have gained from the history of science. While students may not change their concepts of science precisely as scientists have, there does appear to be enough similarity to expect the parallel to be illuminating.

We can sketch our views on conceptual change by contrasting them with classical empiricism. Among early empiricist philosophers such as Locke or Hume, the notion that the mind is a blank tablet to be written on by experience was a compelling metaphor. This metaphor performed several roles. It emphasized the role of experience in knowledge, and it focused attention on the question of how the bits and pieces of experience are pasted together into knowledge of the world. It also diminished interest in the role of prior knowledge or prior ideas on current knowledge acquisition. If the mind is a blank tablet, it must be capable of learning directly from experience without the aid of prior conceptions. The metaphor of a blank tablet requires that at the outset of learning there be no prior conceptions.

We understand conceptual change in direct opposition to the classical form of empiricism. Individuals approach any inquiry with a prior set of concepts. The nature of these concepts significantly determines what is learned and how it is learned. Moreover, learning is not just a matter of adding to one's store of concepts. It transforms them in some way. Neither the learning of an individual nor the production of new knowledge by an intellectual profession is the mere accumulation of new facts. It is the transformation of current knowledge.

Two commitments are important in understanding this view of learning. The first is the learning and the production of new knowledge are similar in that they are both rational activities. They involve making judgments about the truth or falsity of ideas on the basis of evidence. The second commitment is that rationality has to do with changing one's mind. The import of this idea stems from the fact that until recently philosophers saw the rationality of a theory as concerned with the relations between the theory and the experimental evidence for it. Most debates in philosophy of science concerned the question of when theories should be held to be verified or falsified by the experimental evidence available. We, however, maintain that the rational acceptance of a new theory is not so much a matter of whether it is corroborated by the evidence but whether it solves the problems generated by its predecessors. Empirical evidence is not irrelevant here, but being rational has primarily to do with how we move from one view to another.

These two claims have surprising mileage in them for understanding science teaching and learning. They suggest that the learner in the classroom is like the scientist working on the forefront of knowledge. Moreover, they suggest that the history and philosophy of science has light to shed on learning. Not every student is a Newton or an Einstein, but every serious student, like every great scientist, is faced with reorganizing old beliefs in the light of new ideas. Understanding learning requires us to understand how conceptions change.

Our views concerning the character of conceptual change draw heavily on the work of such individuals as Thomas Kuhn (1970), Imre Lakatos (1970) and Stephen Toulmin (1972). We assume a general familiarity on the part of the reader with the work of these people. We do, however, wish to note

Downloaded by [University of Iowa Libraries] at 06:35 14 March 2015

briefly the facets of their views which we see as important to our view of conceptual change in learning.

First, we emphasize the importance of substantive concepts in inquiry and learning. Modern empiricist theories explicate scientific method by means of the concepts of formal logic. They view the problems of philosophy of science largely as reconstructing the formal logical relations between scientific theories and observation statements. We wish to insist, in contrast, on the central role that substantive scientific commitments have in scientific thought. What scientists currently believe has a considerable impact on the research problems they select, on how they conceive the problems, on the methods they use to investigate them, and on what they consider to be adequate problem solutions. To understand scientific thought, it is necessary to understand the content of scientific belief systems and how they function in inquiry.

Second, we believe that a distinction of the sort that Kuhn draws between normal and revolutionary science is important for an understanding of how belief systems affect scientific thought. Typically, scientific thought is dominated by a set of basic assumptions or beliefs which generate questions and problems, but which are not themselves at issue in the research. Thought is rather directed toward applying, extending, and elaborating these enduring conceptual structures. At other times, however, these more central concepts are themselves at issue. In these cases a science will be forced to examine some of its central beliefs and will, as a result, be less sure of its direction.

Finally, we emphasize that science does not grow simply by the addition of new facts. Nor does it grow simply by replacing old inadequate theories by new more adequate ones. Instead, new scientific ideas and discoveries interact with and affect the content and organization of current concepts. The inquirer is not simply adding or subtracting pieces of inventory in his conceptual warehouse. Instead, he faces the task of establishing a reflective equilibrium between new ideas, facts, and discoveries and his current set of concepts. Understanding scientific thought is a matter of understanding how such adjustments in set of concepts occur and how reflective equilibrium is achieved.

Our research has focused on large-scale conceptual changes analogous to Kuhnian 'paradigm changes' or Lakatosian 'shifts between research programmes'. Our reasons for this are two. First, there has been a good deal of work in education which can be regarded as concerned with the sorts of conceptual changes which are analogous to Kuhn's normal science. Several authors have emphasized the role of general concepts in learning the specific claims of a field (Bruner 1966, Ausubel 1968). These views, however, often lack a reasonable account of how such central concepts are acquired. Often there is no account at all or a retreat to an empiricist view of learning (Strike and Posner 1976). Somehow it seems that prior concepts are necessary for learning all except the most important ideas. For them the mind becomes a blank tablet.

Second, large scale conceptual changes are more philosophically problematic and thus more philosophically interesting. Indeed, Kuhn's tendency to relativism concerning revolutionary science and the tendency of

Downloaded by [University of Iowa Libraries] at 06:35 14 March 2015

educators to lapse into empiricist accounts in dealing with central concepts have some interesting similarities. Both are instances of the difficulty of accounting for how a new concept is judged or acquired, when there are no relevant additional concepts to which it can be related. Kuhn generates this problem by putting so much into a paradigm that, when the paradigm is in doubt, it becomes difficult to identify any firm intellectual ground on which one can stand to make a reasoned judgment. The educational version of this problem is 'the problem of the first case'. It is common for educators to hold that new concepts are rendered meaningful by exhibiting their relations to other, more central concepts. But it seems that this account cannot be true for all concepts. There must be a first case and this must be a concept learned without reference to any other. Both are suggestive of an empiricist solution. There must be some things we can come to know without knowing anything else. Perhaps we do start with a blank tablet.

We are, however, disinclined toward such a conclusion. When students are asked to revise some central concept, normally there are available intellectual resources which enable them to begin to judge new alternatives. Most important among such resources are the difficulties generated by their current views. That a new concept can solve the problems generated by its predecessor is a major factor in its appraisal. Sometimes students are introduced to a concept which is not only unfamiliar but which concerns an unfamiliar range of facts and problems. Even in such cases they will have concepts available to them dealing with other types of phenomena. These concepts can be used to conduct at least an initial appraisal of such new ideas (*see* Toulmin 1972, for a discussion of this point). Students never approach any concepts, new or otherwise, with heads entirely devoid of cognitive resources for their appraisal. Indeed, we believe there is considerable evidence that even newborn infants come into the world with cognitive equipment which they use to sort through William James' 'blooming buzzing confusion' of experience (*see* Kiel 1981). We thus see the issue as one of identifying the available intellectual resources by means of which a learner can assess a new central concept. Identifying these conceptual components of large scale conceptual change is the basic question of our work.

We have elected to distinguish between large-scale and small-scale conceptual changes by means of the labels 'accommodation' and 'assimilation'. An accommodation occurs when an individual must modify some central or basic concept. Paradigm shifts or changes of one's conceptual hard-core would count as instances of an accommodation.

This selection of terminology involves several commitments. We wish to reject both extreme evolutionary and extreme revolutionary views. In effect, this means that while we do agree with Kuhn and Lakatos that some concepts are more central to a given scientific enterprise than others, we reject the notion that there are cases where all of our concepts are in doubt simultaneously. There is never a point at which no concepts are available to provide some criteria for the selection of new ones. These ideas suggest that the distinction between accommodation and assimilation is both relative to the domain of concerns looked at, and a matter of degree rather than kind. These commitments also allow us to understand accommodation in a way that does not generate the problem of the first case and which poses the

Downloaded by [University of Iowa Libraries] at 06:35 14 March 2015

question of the character of the cognitive content which functions in the process of accommodation.

Conditions for conceptual change

Since our interests concern accommodation, it is in order to suggest some of the hypotheses we are entertaining concerning the process of accommodation. Most of these are suggested by work in the history and philosophy of science, but, we believe, can be applied to the learning situation as well. First, consider the question 'Under what conditions is an accommodation likely to occur?' Standard accounts suggest some of the following:

(*a*) *There must be dissatisfaction with existing conceptions*

Scientists are unlikely to make major changes in their concepts until they believe that less radical changes will not work. Thus, before an accommodation will occur, it is reasonable to suppose that an individual must have collected a store of unsolved puzzles or anomalies and lost faith in the capacity of his current concepts to solve these problems.

(*b*) *A new conception must be intelligible*

The individual must be able to grasp how experience can be structured by a new concept sufficiently to explore the possibilities inherent in it. Writers often stress the importance of analogies or metaphors in lending initial meaning and intelligibility to new concepts.

(*c*) *A new conception must appear initially plausible*

Any new concept adopted must at least appear to have the capacity to solve the problems generated by its predecessors. Otherwise it will not appear a plausible choice. Plausibility is also a result of consistency of the concept with other knowledge. A new idea in, say astronomy, is less likely to be accepted if it is inconsistent with current physical knowledge, or if it simply has no clear physical account. Physical scientists prior to the twentieth century were, for example, reluctant to accept what geologists were claiming about the age of the world, since they had no theory which would allow the sun to provide energy for that period of time.

(*d*) *A new concept should suggest the possibility of a fruitful research programme*

The new conception must do more than solve its predecessors' difficulties. It should have the potential to be extended, to open up new areas of inquiry.

Features of a conceptual ecology

The foregoing comments suggest a second question. If new concepts are selected in the context of more enduring concepts, what sorts of concept are these? If we can adopt a term from Toulmin's evolutionary theory, what are the features of a 'conceptual ecology' which make a conceptual change possible? Some of the following should be of interest:

Downloaded by [University of Iowa Libraries] at 06:35 14 March 2015

1. *Anomalies*: The character of the specific failures of a given idea is an important part of the ecology which selects its successor.
2. *Analogies and metaphors*: These can serve to suggest new ideas and to make them intelligible (Belth 1977, Black 1962, Ortony 1975).
3. *Epistemological commitments*:
   (i) Explanatory ideals—most fields have some subject matter-specific views concerning what counts as a successful explanation in the field.
   (ii) General views about the character of knowledge—some standards for successful knowledge such as elegance, economy, parsimony, and not being *ad hoc* seem subject matter-neutral (Lakatos 1970, Elkana 1977).
4. *Metaphysical beliefs and concepts*:
   (i) Metaphysical beliefs about science—beliefs concerning the extent of orderliness, symmetry, or non-randomness of the universe are often important in scientific work and result in epistemological views which in turn can select or reject particular kinds of explanations. Beliefs about the relations between science and commonplace experience are also important here (Burtt 1932).
   (ii) Metaphysical concepts of science—specific scientific concepts often have a metaphysical quality, in that they are beliefs about the ultimate nature of the universe and are immune from direct empirical refutation. A belief in absolute space or time is an example (Burtt 1932).
5. *Other knowledge*:
   (i) Knowledge in other fields.
   (ii) Competing concepts—one condition for the selection of a new concept is that it should appear to have more promise than its competitors.

Two comments on this list of possible components of an ecology may be helpful. These kinds of features are understood as governing the initial selection of a new concept. They are much less concerned with its ultimate justification. Thus, we are concerned with the generation of new ideas, with the context of discovery, rather than with the context of verification. Second, the particular components of this list which are relevant to a given conceptual change will likely vary depending on the particular issue, and whether the individual is a professional scientist researching a new area or a novice mastering a new field. Scientists do their work in a conceptual ecology, and students study the ideas of scientists in their own conceptual ecology. One should not uncritically assume the two ecologies to be the same.

This raises the final question of this section. Why would one suppose that one can learn something about how students learn science by studying the history and philosophy of science? Here we need to suggest what we do and do not assume about the relations between a scientist investigating a new area, and the student subsequently mastering what that scientist discovered. What we do assume is that the student and the scientist share a common problem. Both have a current set of concepts. If a new concept is going to be

Downloaded by [University of Iowa Libraries] at 06:35 14 March 2015

learned by the student or adopted by the scientist, it is necessary that a rational path be seen leading from current concepts to new ones. Both must see that a change in one or more concepts is needed and that this change is justified in terms of the evidence and the remaining store of concepts. What we do not assume is that the path the student must follow from old concepts to new ones is the same as the path from one historical scientific period to its successor. The development of students' concepts need not recapitulate the scientific history.

We believe we can derive two sorts of belief with regard to science learning from the history and philosophy of science. We can discover patterns of rational conceptual development. The views of people such as Kuhn, Lakatos, Toulmin, and Holton provide a repertory of ideas which can be used to investigate conceptual change in students. Also, the history of science can provide hypotheses concerning the particular concepts which may be involved in a student's conceptual change. Thus, while students may not recapitulate the history of a discipline and their concepts may not change precisely as concepts in scientific change, there does appear to be sufficient similarity such that the history and philosophy of science can illuminate science learning. Perhaps that is what one would expect if one views learning as a rational activity.

Downloaded by [University of Iowa Libraries] at 06:35 14 March 2015

Conceptual change: implications for educational practice

The epistemological views we have sketched suggest that instruction designed to promote rational conceptual change might have two primary emphases. First, anomalies appear to have a crucial role in rational conceptual change. Old theories are rejected because they generate problems they cannot solve. New theories are adopted because they solve such problems. If students are to accommodate some new theory, they need to see what problems that theory can solve. We hasten to add that the problems a new conception solves are not the same thing as the data the conception explains. A problem is an inconsistency among current concepts which may (but need not) arise because of some new piece of data. For a student to accommodate a new idea, it must appear to solve problems embedded in prior concepts of the scientific community or of the students—preferably the latter.

Second, we have argued that new conceptions are judged in the context of a conceptual ecology, and that metaphysical and epistemological concepts may be especially important components of that ecology. One of the surprises we have encountered in our research is the extent to which such metaphysical and epistemological assumptions inhibit or promote accommodation. This seems to be the case both in the history of science (Burtt 1932) and for students learning science (Posner *et al.* 1982).

There is considerable evidence to suggest, for example, that the work of Copernicus, Kepler, and Galileo was substantially motivated by a revival of Neo-Platonism which viewed the cosmos as intrinsically mathematical and thus as describable by mathematical formulae. Moreover, in some interviewing we have done of physics students struggling to accommodate Einstein's special theory of relativity, we have found that initial concepts of space and

time and of the character of science and scientific evidence make a considerable difference in how special relativity is learned (Hewson 1980, Posner *et al.* 1982). Some students appear almost driven to reconstruct special relativity in Newtonian terms. Also, students often fail to see anomalous results as problematic in ways that seem rooted in epistemological beliefs which lack strong demands for consistency or symmetry among theory and phenomena.

Such observations tend to highlight the fact that the students' conceptual ecology can differ from that of both the scientists who made the discoveries about which the student is expected to learn, and from currently practising scientists. Few students, we suspect, are Neo-Platonists. Thus, they will not find the arguments which persuaded Copernicus or Kepler especially convincing. Moreover, many students may lack the scientific knowledge or the epistemological concepts necessary to recognize as anomalous the things which professional scientists will see as problematic. We thus need to ask what can be done to enable science students to accommodate new conceptions on a rational basis. Instruction needs to focus on filling in the gaps in a student's conceptual ecology—on building a rational bridge from current concepts to new ones.

Downloaded by [University of Iowa Libraries] at 06:35 14 March 2015

Curricular objectives

Our discussion of the critical role played by the students' fundamental assumptions about the world and about their knowledge of the world raises serious questions about the objectives of science courses. If the conceptual change process is to be rationally based, students will need to be immunized against the kind of inevitable indoctrination that occurs when neither the teacher nor the student is aware of his own fundamental assumptions, much less those implied by the science they are teaching and learning. Conversely, much of the conceptual ecology required for rational conceptual change appears to consist of deeply rooted metaphysical and epistemological concepts. Thus, the preceding discussion leads to some scepticism regarding the feasibility of fundamental conceptual change in science courses. Perhaps we should aim only at making new theories intelligible to students. However, if we are interested in students' internalizing new conceptions whose epistemological and metaphysical foundations conflict with those currently held by the student, then we might wish to add two dimensions to our existing science courses. We might wish to aim at an awareness of the epistemological and metaphysical foundation of modern science, perhaps through an historical approach. We might also aim at an awareness by students of their own epistemological and metaphysical assumptions and an ability to juxtapose their assumptions with those implicit in scientific theory.

Content

If we aim to produce rationally-based conceptual change in students, then according to what we have said thus far, the content of science courses should be such that it renders scientific theory intelligible, plausible and fruitful. This general requirement suggests two points.

First, anomalies should be employed. It is important to use anomalies which students can experience as anomalous, given either their current knowledge and beliefs or background material supplied by the teacher. Anomalies that are historically valid will not necessarily work. In such cases, anomalies will have to be selected or constructed so that they are easy for students to understand and conflict with the students' preconceptions. This approach, of course, assumes some knowledge of the students' preconceptions.

Second, models, prototypical examples and analogies should be carefully chosen to make a new conception more intelligible and plausible. Care must also be exercised so as to avoid over-generalization and other misuses of this instrumental content.

Downloaded by [University of Iowa Libraries] at 06:35 14 March 2015

Teaching strategies

When aimed at recall and assimilation, teaching typically includes clarifying content presented in texts, explaining solutions to problems, demonstrating principles, providing laboratory exercises, and testing for recall of facts and ability to apply knowledge to problems. If teaching were to be aimed at accommodation, lectures, demonstrations, problems and labs might also be used to create cognitive conflicts in students and to help students experience anomalies.

Further, accommodation is more likely to occur if instruction can be organized so that teachers can spend a substantial portion of their time in diagnosing and correcting errors in student thinking.

Finally, techniques are needed to help teachers determine the students' preconceptions and to help track the process of conceptual change. Clinical interviews (Posner and Gertzog 1982), analyses of student errors (Resnick 1980), analyses of students' 'thinking aloud protocols' (Simon and Simon 1979), and questionnaires (Strike *et al.* 1981) appear promising.

Teacher role

If accommodation is to occur, the teacher might have to be not only a clarifier of ideas and presenter of information, but also an adversary and a model of scientific thought.

We mean adversary in the sense of a Socratic tutor, the adversarial role directed at the student's ideas, not at the student as a person. In this role, the teacher relentlessly confronts the students with the problems arising from their attempts to assimilate, rather than to accommodate, new conceptions.

As a model of scientific thought, the teacher insists on consistency among beliefs, is sceptical of excessive '*ad hocness*' in theories, and critically analyzes empirical results in order to determine whether any discrepancies between results and theory constitute an anomaly or are in reasonable agreement with theory.

These ideas obviously fall rather short of a detailed plan for teaching science. They are best viewed as the structure of a research programme. They indicate some of the questions to be asked in developing curriculum and instructional techniques for science teaching. Their virtue as a research programme is that they are rooted in a view of learning which makes

epistemological sense. Perhaps educators have lost sight of the fact that one criterion for judging views of teaching and learning is that they should be commensurable with our best understanding of the character of knowledge and inquiry.

References

AUSUBEL, D. P., 1968, *Educational Psychology: A Cognitive View* (Holt, Rinehart and Winston: New York).

BELTH, M., 1970, *The Process of Thinking* (David McKay: New York).

BLACK, M., 1962, *Models and Metaphors* (Cornell University Press: Ithaca, New York).

BRUNER, J., 1966, *Toward a Theory of Instruction* (Harvard University Press: Cambridge).

BURTT, E. A., 1932, *The Metaphysical Foundations of Modern Science*, revised edition (Doubleday and Company, Doubleday Anchor Books: Garden City, New Jersey).

ELKANA, Y., 1970, The historical roots of modern physics. In *Italian Physical Society, Proceedings of the International School of Physics: History of Twentieth Century Physics* edited by C. Weinter (Academic Press: New York).

HEWSON, P. J., 1980, Learning and teaching science. *South African Journal of Science*, Vol. 76, pp. 397–403.

KIEL, F. C., 1981, Constraints on knowledge and cognitive development, *Psychological Review*, Vol. 88, pp. 197–227.

KUHN, T., 1970, *The Structure of Scientific Revolutions*, 2nd edition (University of Chicago Press: Chicago).

LAKATOS, I., 1977, Scientific research programmes. In *Italian Physical Society, Proceedings of the International School of Physics: History of Twentieth Century Physics*, edited by C. Weinter (Academic Press: New York).

ORTONY, A., 1975, Why metaphors are necessary and not just nice. *Educational Theory*, Vol. 25, pp. 45–53.

POSNER, G. J., STRIKE, K. A., HEWSON, P. W., and GERTZOG, W. A., 1982, Accommodation of a scientific conception: Toward a theory of conceptual change, *Science Education*, Vol. 66, pp. 211–228.

POSNER, G. J., and GERTZOG, W. A., 1982, The clinical interview and the measurement of conceptual change, *Science Education*, Vol. 66, pp. 195–209.

RESNICK, L., 1980, The role of invention in the development of mathematical competency. In *Developmental Models of Thinking*, edited by R. H. Kluwe and H. Spada (Academic Press: New York).

SIMON, D. P. and SIMON, H. A., 1979, A tale of two protocols. In *Cognitive Process Instruction: Research on Teaching Thinking Skills*, edited by J. Lochhead and J. Clement (The Franklin Press: Philadelphia).

STRIKE, K. A., GERTZOG, W. A., HEWSON, P. J., HOAGLAND, G., NUSSBAUM, J. and POSNER, G. J., 1981, *The Cornell Assessment of Scientific Beliefs (CASB): A report on questionnaire development*. Research Report No. 10, Curriculum Series (Department of Education, Cornell University: Ithaca, New York).

STRIKE, K. A., and POSNER, G. J., 1976, Epistemological perspectives on conceptions of curriculum organization and learning. In *Review of Research in Education: 4*, edited by Lee Shulman (F. E. Peacock: Itasca, Illinois).

TOULMIN, S., 1972, *Human Understanding* (Princeton University Press: Princeton).

Downloaded by [University of Iowa Libraries] at 06:35 14 March 2015